

Evaluating Perfect Square and Cube Roots

Perfect square roots, perfect cube roots, and roots inside larger expressions

Directions

- Every root on this sheet comes out exact — no calculator, no decimals.
 - $\sqrt{\quad}$ asks: what number times itself gives the number inside? $\sqrt[3]{\quad}$ asks: what number used as a factor three times gives it?
 - Write the multiplication that proves your root, then put the value on the answer line.
-

1. Evaluate $\sqrt{49}$.

Answer: _____

2. Evaluate $\sqrt[3]{27}$.

Answer: _____

3. Evaluate $\sqrt{144}$.

Answer: _____

4. Evaluate $\sqrt[3]{64}$.

Answer: _____

5. Evaluate $\sqrt{81} + \sqrt[3]{8}$.

Answer: _____

6. Evaluate $\sqrt{225}$.

Answer: _____

7. Evaluate $\sqrt[3]{125}$.

Answer: _____

8. Evaluate $3\sqrt[3]{216} - \sqrt{100}$.

Answer: _____

9. A square patio covers exactly 169 square feet of ground. How long is one side of the patio?

Answer: _____ ft

10. A solid cube-shaped crate holds exactly 512 cubic inches. How long is one edge of the crate?

Answer: _____ in

$$\sqrt{64} \quad \underline{\hspace{1cm}} \quad \sqrt[3]{216}$$

(b) Explain what Kara did wrong. Name the root she took instead, and show a multiplication check that settles it.